La Scuola d'Italia / Dr. Huson / IB Math
17 April 2026 Markscheme

**6.8 Quiz: Polynomials — Markscheme**

1. [Maximum mark: 7]

Let $f(x) = x^2 - 4x - 12$.

**(a)** Write $f(x)$ in factored form $(x-a)(x-b)$. [2]

$f(x) = (x-6)(x-(-2)) = (x-6)(x+2)$ **(M1)(A1)**

*Accept* $(x+2)(x-6)$.

**(b)** Write down the $x$-intercepts. [1]

$x = 6$ and $x = -2$ **(A1)**

**(c)** Write down the $y$-intercept. [1]

$f(0) = -12$, so the $y$-intercept is $(0, -12)$. **(A1)**

**(d)** Write $f(x)$ in vertex form $(x-h)^2 + k$. [1]

$f(x) = (x-2)^2 - 16$ **(A1)**

*i.e.* $h = 2$, $k = -16$.

**(e)** Find the coordinates of the vertex. [2]

Axis of symmetry: $x = \dfrac{-2+6}{2} = 2$ **(A1)**

$f(2) = 4 - 8 - 12 = -16$

Vertex: $(2, -16)$ **(A1)**

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2. [Maximum mark: 7]

Consider the function $g(x) = -(x-3)^2 + 16$.

**(a)** Write down the coordinates of the vertex. [1]

Vertex: $(3, 16)$ **(A1)**

**(b)** Write down the equation of the axis of symmetry. [1]

$x = 3$ **(A1)**

**(c)** Show that $g(x) = -x^2 + 6x + 7$. [2]

$g(x) = -(x-3)^2 + 16 = -(x^2 - 6x + 9) + 16$ **(M1)**

$= -x^2 + 6x - 9 + 16 = -x^2 + 6x + 7$ **(A1)(AG)**

**(d)** Solve $g(x) = 0$ to find the $x$-intercepts. [3]

$-x^2 + 6x + 7 = 0$ **(M1)**

$x^2 - 6x - 7 = 0$

$(x - 7)(x + 1) = 0$ **(A1)**

$x = 7$ or $x = -1$ **(A1)**

3. [Maximum mark: 7]

The line $y = 3x$ and the parabola $y = x^2 - 4$.

**(a)** Show that the $x$-coordinates of intersection satisfy $x^2 - 3x - 4 = 0$. [2]

At intersection: $x^2 - 4 = 3x$ **(M1)**

$x^2 - 3x - 4 = 0$ **(A1)(AG)**

**(b)** Show that $x^2 - 3x - 4 = 0$ has two distinct real roots. [1]

$\Delta = (-3)^2 - 4(1)(-4) = 9 + 16 = 25 > 0$ **(A1)**

Since $\Delta > 0$, there are two distinct real roots.

**(c)** Factorise $x^2 - 3x - 4$. [1]

$(x - 4)(x + 1)$ **(A1)**

**(d)** Find the $x$-coordinates of the points of intersection. [1]

$x = 4$ and $x = -1$ **(A1)**

**(e)** Find the coordinates of both points of intersection. [2]

$x = 4$: $y = 3(4) = 12 \implies (4, 12)$ **(A1)**

$x = -1$: $y = 3(-1) = -3 \implies (-1, -3)$ **(A1)**

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4. [Maximum mark: 7]

The equation $3x^2 - kx + 12 = 0$ has two equal roots.

**(a)** Write down the discriminant in terms of $k$. [1]

$\Delta = (-k)^2 - 4(3)(12) = k^2 - 144$ **(A1)**

**(b)** Find the values of $k$. [2]

Equal roots: $\Delta = 0$

$k^2 - 144 = 0 \implies k^2 = 144$ **(M1)**

$k = \pm 12$ **(A1)**

**(c)** For $k = 12$, find the repeated root. [2]

$3x^2 - 12x + 12 = 0 \implies x^2 - 4x + 4 = 0$ **(M1)**

$(x - 2)^2 = 0 \implies x = 2$ **(A1)**

**(d)** State the values of $k$ for which the equation has no real roots. [2]

$\Delta < 0 \implies k^2 - 144 < 0 \implies k^2 < 144$ **(M1)**

$-12 < k < 12$ **(A1)**